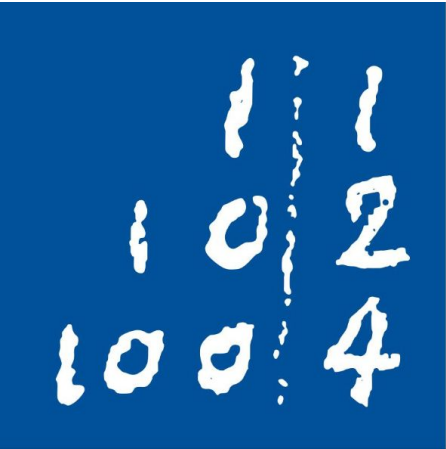


How Far Back Must the Gradient Reach?

Truncated BPTT length and memory in PPO on MiniGrid

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Poster presentation in context of the Reinforcement Learning lecture



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1 TL;DR

A recurrent policy can only use a memory that its **gradient** has taught it to write. One PPO pipeline, fixed; the only thing varied is how far back that gradient may reach — the TBPTT window L .

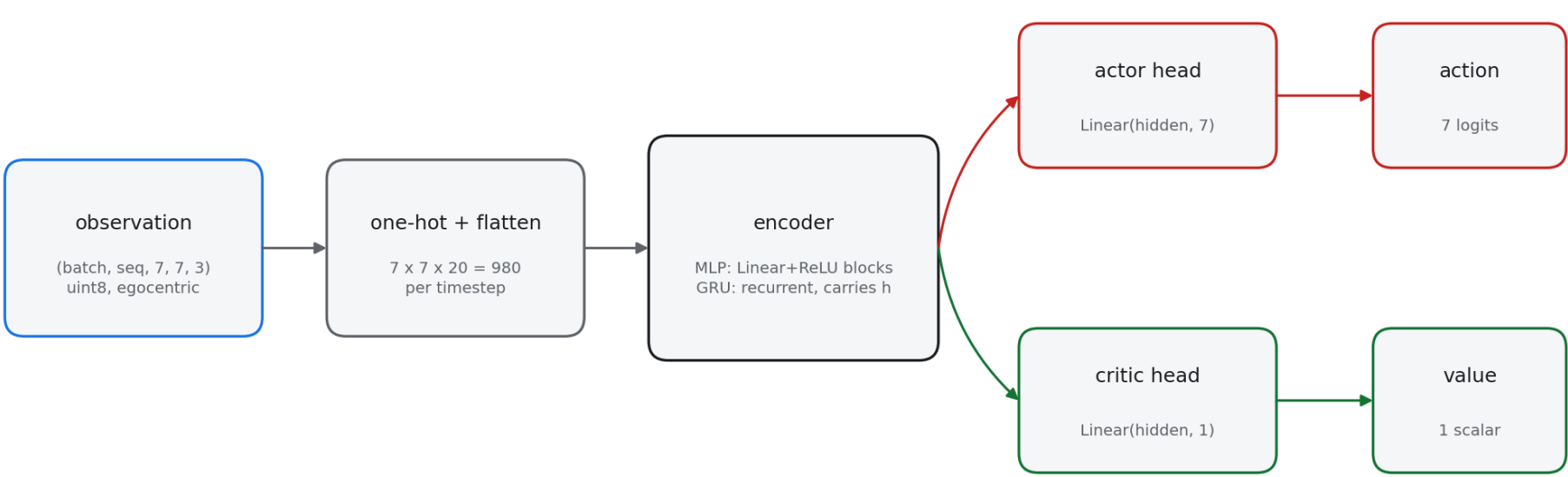
The threshold is sharp. $L \geq 8$ solves the task on every seed; $L \leq 4$ never beats a coin flip. Reaching further costs VRAM, not time.

1.000	0.500	8	8×
success at $L = 8$ and full BPTT	success at $L = 4$ (chance)	steps of reach suffice (4 do not)	the VRAM it costs (the time: none)

5 studies $\times$ 50 trials $\times$ 5 seeds = **250 training runs**, $\approx$ 236 GPU-hours on one RTX 3060 Laptop (12 GB).

3 Approach

Network



$7 \times 7 \times 3$ view $\rightarrow$ one-hot (980) $\rightarrow$ encoder $\rightarrow$ linear actor + linear critic. The encoder is the **only** difference between arms: an MLP (memoryless) or a single GRU layer carrying h forward.

The knob under study

Before every update the rollout is cut at each **episode boundary and every L steps**. A chunk starting mid-episode is seeded with the hidden state the rollout recorded there, so the forward pass still sees the whole episode and **only the backward pass is shortened** — every GRU arm carries identical information forward.

One PPO iteration

- collect $W = 32$ workers $\times$ $T = 361$ steps, recording $(o, a, \log \pi, V, r, d, h)$
- GAE over the **whole** rollout, before any split; standardise $\hat{A}$
- cut into chunks of $\leq L$ steps, each seeded with its stored h**
- 3 shuffled epochs of the clipped PPO loss; padding never enters it

$L = \infty$ (full BPTT) cuts at episode boundaries only; a smaller L means more, shorter sequences — $\lceil T/L \rceil$ of them.

Tuning protocol

- One Optuna study **per arm**: 50 TPE trials over **identical** ranges — no arm is judged on another's settings.
- Each trial trains the same 5 seeds for 1000 iterations and is scored **mean – std across seeds** on 50 held-out mazes.
- Nothing that buys compute is searchable; the minibatch size is set by a VRAM probe, so every arm is measured at the **same memory budget**.

6 Limitations & Future Work

One task, one architecture, one algorithm. $L \in \{5, 6, 7\}$ was never run, so the threshold is **bracketed, not located**, and five seeds separate 0.5 from 1.0 safely but support no finer claim.

- Locate the threshold.** Sweep $L \in \{5, 6, 7\}$ and set the corridor length directly instead of leaving it to the environment — a causal test of the gap account.
- Add R2D2-style burn-in.** Does a short window then behave like a long one? If so the threshold is about optimisation, not the gradient path.
- Repeat with an LSTM and a small transformer**, where the analogous knob is the attention window.

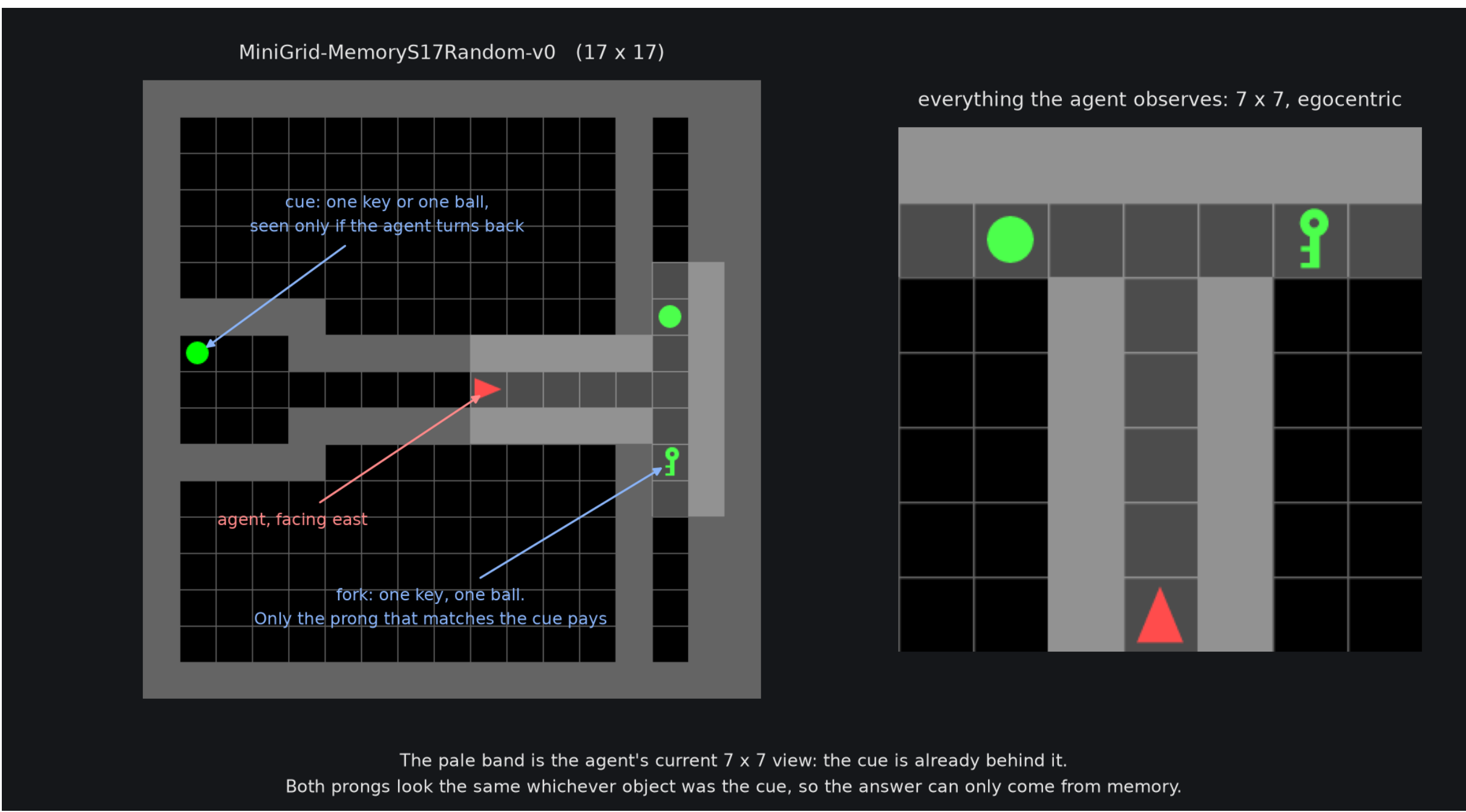
References. Hausknecht & Stone 2015 · Kapturowski et al. 2019 (R2D2) · Ni et al. 2022 · Schulman et al. 2017 (PPO) · Chevalier-Boisvert et al. 2023 (MiniGrid)

2 Motivation & Problem Setting

In a POMDP the optimal action depends on information that has left the observation. The usual remedy is a recurrent policy; the usual concession is to **truncate the backward pass to L steps**, a value inherited in silence from whatever codebase one starts from.

MiniGrid-MemoryS17Random-v0 makes it measurable. A cue visible only in the west room decides which prong of the T-junction at the far end pays, and the 7×7 egocentric view makes both prongs identical: a **memoryless policy is capped at 0.5**, so the gap from 0.5 to 1.0 is exactly the memory a run has acquired.

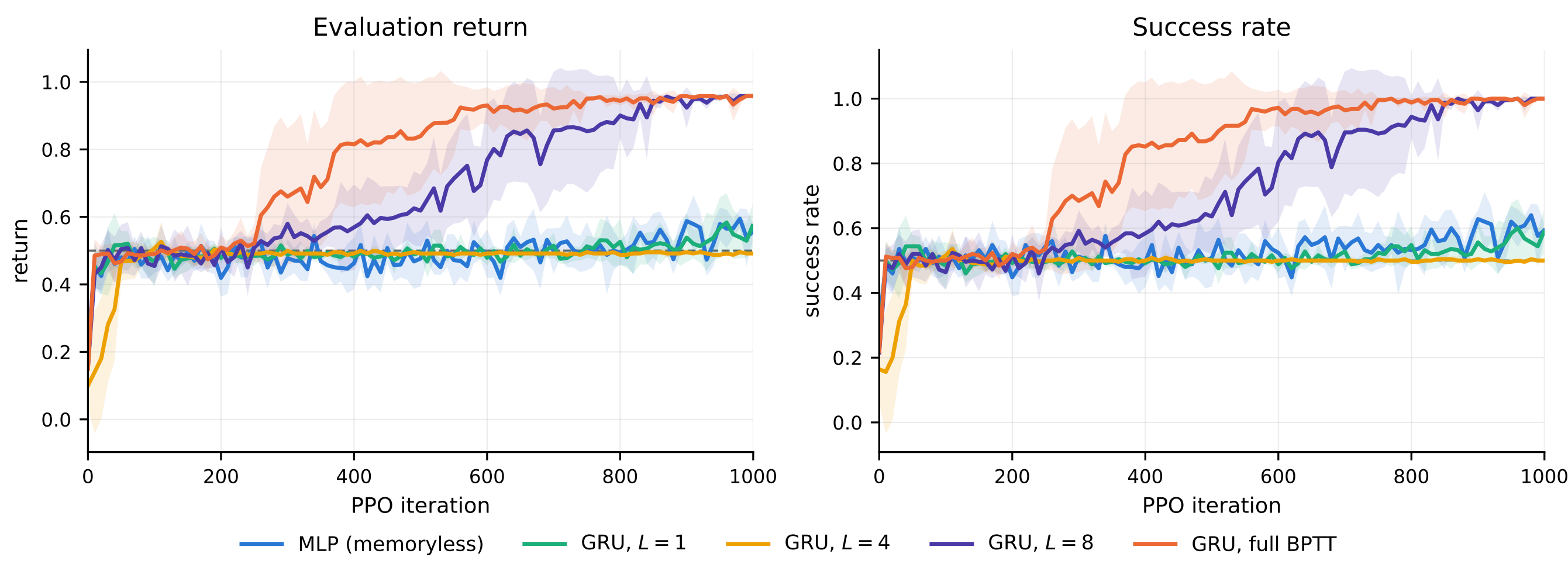
- Q1** — How far back must the gradient reach? Carrying h forward is free at any L , so is a short window enough to **learn what to store?**
- Q2** — What does that reach cost in time and in VRAM? A shorter window is intuitively cheaper — we check whether it is.



4 Results

arm	return	success rate	steps	minibatch	time
MLP (memoryless)	0.556 ± 0.029	0.592 ± 0.027	24.3	1024	1.3 h
GRU, $L = 1$	0.573 ± 0.064	0.588 ± 0.068	9.8	4096	2.6 h
GRU, $L = 4$	0.492 ± 0.001	0.500 ± 0.000	7.0	4096	1.6 h
GRU, $L = 8$	0.958 ± 0.004	1.000 ± 0.000	16.8	4096	2.0 h
GRU, full BPTT	0.958 ± 0.002	1.000 ± 0.000	16.8	512	1.3 h

Winning trial per study, averaged over its five seeds; $\pm$ is the spread across seeds. No setting produced an intermediate score.



Mean over 5 seeds, band at ± 1 std, dashed line at chance. The two solving arms separate before iteration 400; the three failing arms never leave the chance line.

5 Key Insights

- Carrying h forward is not what matters.** Every GRU arm does it identically, including the ones that fail. What separates them is whether the **gradient** reaches back to the observation that filled h — and reaching past 8 adds nothing.
- What has to fit inside L is the gap, not the episode.** The gap is the number of actions between the agent's last sighting of the cue and its choice at the fork. By BFS over the environment's own visibility model, a gap ≤ 8 is attainable in **62 %** of mazes but a gap ≤ 4 in only **26 %**; replaying the checkpoints, cue and decision share a chunk in 16 % of the $L = 8$ agent's episodes and in **0 of 50** of the $L = 4$ agent's.
- The failing arms are not partially remembering.** At 7.0 steps the $L = 4$ arm dashes straight to the fork and always picks the same side — exactly 25/50 here. Its across-seed std of 0.000 is **arithmetic, not convergence**.
- The reach costs memory, not time.** Full BPTT is the **fastest** GRU arm (1.3 h vs 2.6 h at $L = 1$) and the earliest (0.95 success at a median of 370 iterations vs 640 for $L = 8$): halving L doubles the padded, launch-bound sequences. What it pays is VRAM — minibatch 512 vs 4096.

Recommendation. On this task: full BPTT, with $L = 8$ as the fallback on a smaller GPU. Below 8 you do not get a cheaper solution — you get the MLP's guessing policy at GRU prices.